Yellow Jacket Swarm Counter Unmanned Aerial System

DIU Counter UAS Sensing for Homeland and Mobile Defense (PROJ00656)
Interstellar Foundry retains commercial IP for the core sensor-fusion algorithm, while granting the DoD Government Purpose Rights to specific interface APIs and ASCII outputs

Project High Level Overview

Role:

- Designed and built the interceptor drone entirely. (Ideation, CAD, manufacturing, controls, testing)
- Now, lead 6-person team where I direct the hardware and manufacturing while integrating with SDEs

Timeline

- Last semester: Physical interceptor
- This semester: Drone Detection System
- Aim to reach TRL 6 by April

Program:

- Georgia Tech Create-X Team 7, Interstellar Foundry

Customers:

- DoW, DIU Open Solicitation, Army FUZE Program, SBIR/STTR

The Threat · Why Now

"Peace cannot be kept by force; it can only be achieved by understanding." — **Einstein**

Low-RCS, Low-Altitude

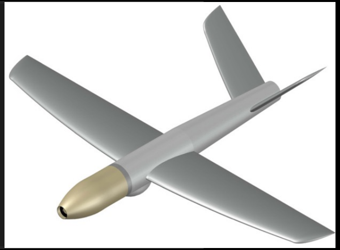
Commercial drones fly under legacy radar thresholds and Small cross-sections evade S-band sensors designed for aircraft.

Sustainment Cost

Current COTS counter-drone solutions cost \$50K–\$500K per node. Cannot scale to protect dispersed DoD infrastructure.

Fixed-Wing Gap

Fixed-wing interceptors lack agility to track quadcopters making abrupt evasive maneuvers at 25+ mph in close engagements.



OUR SOLUTION

Sub-\$1,000 modular ground station

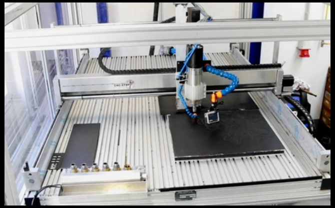
- NVIDIA Jetson Nano edge computing
- FMCW radar + CV sensor fusion
- Kalman filter → eliminates false positives
- 3× autonomous quadcopter interceptors
- Waterjet CF frames → < \$280/unit
- < 15 min deployment on ISV/JLTV

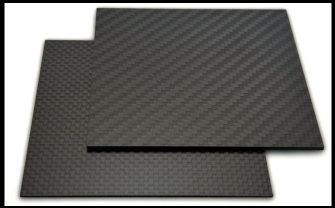
Georgia Tech Create-X · DIU PROJ00656

UAV Interceptor · Manufacturing Approach

HARDWARE SPECS

Frame	450mm quad · waterjet-cut carbon fiber · 3D-printed mounts
Motors	2204 2300KV brushless x 4 per airframe
Props	5" balanced props · optimized for 80+ km/h intercept
Flight Ctrl	Matek F405 running ArduCopter firmware
Comms	ESP32 WiFi MAVLink telemetry to ground station
Cost	~$280/unit · ~$1,617 total for 3 swarm C-UAS
Weight	< 10 lb deployable ground station
Deploy	< 15 min setup





DESIGN FOR MANUFACTURABILITY

Waterjet-Cut CF Frames

2D DXF files → waterjet table. Minimal downtime. Consistent tolerance ±0.005"

3D-Printed Mounts & Arms

PETG/ABS mounts for motors, ESCs, sensors. Rapidly iterate design.

COTS Avionics Stack

Matek F405 + ESP3: proven, in-stock, programmable. No defense foundry lead times. ATO-streamlined.

Cost Scalability

Tier 1: ~$280/unit MVP.
Tier 2 (6-25): Low-Rate Production.
Tier 3 (26-100+): mass manufacturing.

Flight Physics · Design Rationale

PROPULSION PHYSICS

Hover condition:

4T = mg | T = (1/2)rho*A*v_i^2

T_total ~120 N, mass=0.280 kg -> T/W = 43.7

Drag at 80 km/h (22 m/s):

F_drag = (1/2)*rho*Cd*A*v^2

= (1/2)(1.2)(0.4)(0.04)(22)^2 = 4.7 N

Kinetic impact energy at intercept:

KE = (1/2)*m*v_rel^2

Head-on closing @ 44 m/s: KE ~ 271 J

Stress in CF frame (motor mount):

sigma_vm = sqrt(sigma^2 + 3*tau^2)

FEA: 48 MPa at mount holes, FS > 12

DESIGN CHOICES

450mm frame selection

Longer moment arm -> torque per RPM gain. Larger props -> efficiency. Tradeoff: agility vs. 250mm race frame. Intercept geometry prioritizes speed over maneuverability.

2204 2300KV motor selection

High KV -> high RPM at 3S -> fast prop speed -> agility. T/W >4 at hover. >75% thrust reserve for intercept acceleration from 0 to intercept < 3 sec.

3K twill CF plate (3mm)

sigma_UTS ~600 MPa. Waterjet-cuttable with +/-0.005" tolerance. Consistent cross-section vs. hand-layup. Safety factor >12 at motor mounts under crash load.

Polycarbonate guards over CF-PLA

CF-PLA: 220N peak load but catastrophic brittle fracture. PC: 150N but plastic deformation. Failure MODE trumps failure LOAD for attritable interceptors. Must survive, not just endure.

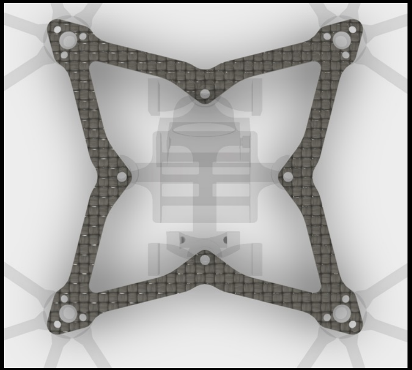
INTERSTELLAR FOUNDRY

Interceptor Design · CAD Modeling

Airframe: 450mm quadcopter configuration. Waterjet-cut carbon fiber main plate (3K twill weave, 3mm thickness). CAD modeled in Fusion 360. 3D-printed PETG motor mounts and camera bracket assembly.

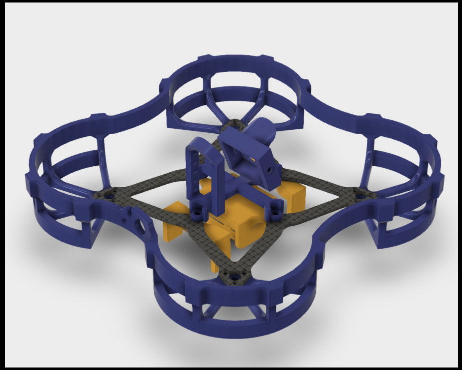
Propulsion: 2204 2300KV brushless motors × 4, 5" balanced propellers, optimized for 80+ km/h intercept.

Flight Control: Matek F405 running ArduCopter 4.3. ESP32 MAVLink telemetry bridge. Coordinated engagement vectors uplinked from ground station via Python API.

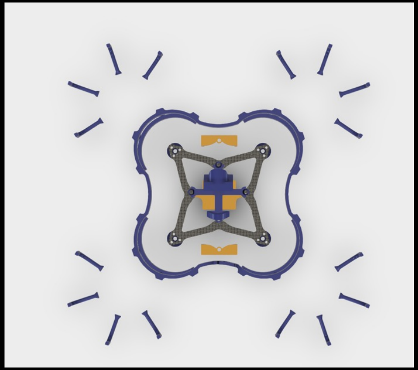


CF Frame (top view)
Waterjet DXF → waterjet table

Payload + Camera Mount
3D-printed PETG



Full Assembly: CF frame (gray) + guards (blue) + mounts (gold)



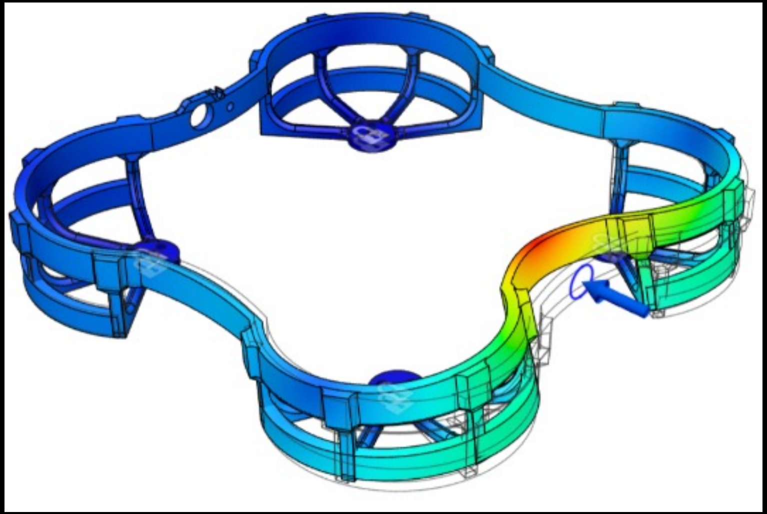
Full Assembly Exploded View

Material Selection · Structural FEA

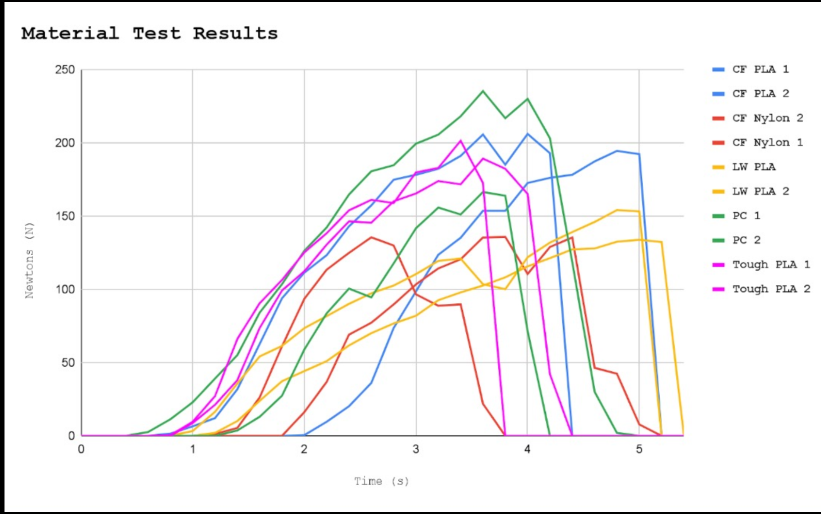
Test Campaign: 10 prop guard specimens across 5 filament types (PETG-CF, CF-PLA, CF-Nylon, LW-PLA, Polycarbonate) loaded to failure under axial compression. Peak load plotted over time to characterize both maximum strength and failure mode (brittle fracture vs. progressive deformation).

Key Finding: CF-PLA achieved peak loads exceeding 220N but fractured catastrophically. Polycarbonate showed the best post-failure behavior (plastic deformation rather than shattering), selected as primary material for guards.

Von Mises Stress: $\sigma_{vm} = \sqrt{\sigma_\theta^2 + \sigma_{ax}^2 - \sigma_\theta \cdot \sigma_{ax}}$



ANSYS FEA: Prop guard structural analysis
Von Mises stress distribution under axial compression load.



Material test result: Force (N) vs. time (s) across 10 specimens.

Physical Build Setup

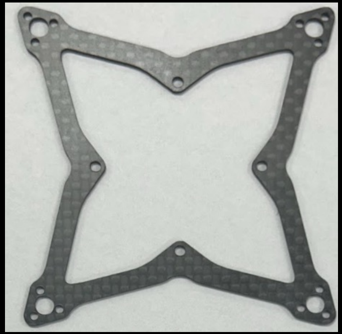
Carbon Fiber Main Plate: 2D DXF exported from Fusion 360 and cut on a waterjet table (±0.005" tolerance).

- **Challenge**: delamination with carbon fiber (piercing in scrap material, using lower pressure)

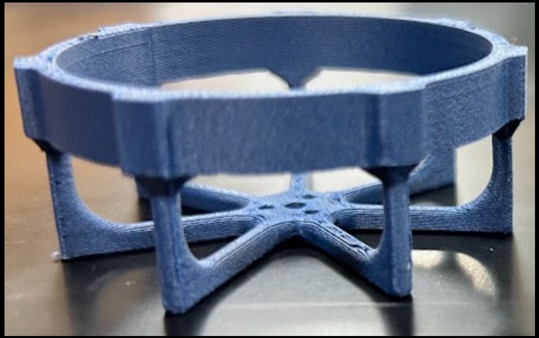
3D-Printed Components: Motor mounts, prop guards, and camera brackets printed in PETG and polycarbonate using Bambu Lab X1 Carbon. Iterated across 5 revs to achieve target budget and impact resilience.

- **Challenge**: Difficulty with polycarbonate (requires high enclosure temps to prevent warping)

Assembly Budget: ~$280/interceptor. Total 3-unit CUAS: ~$1,617.



Waterjet CF frame (physical part)

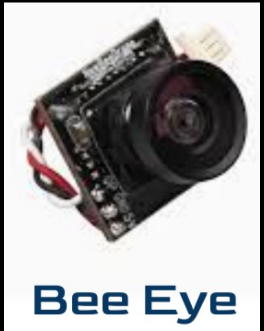


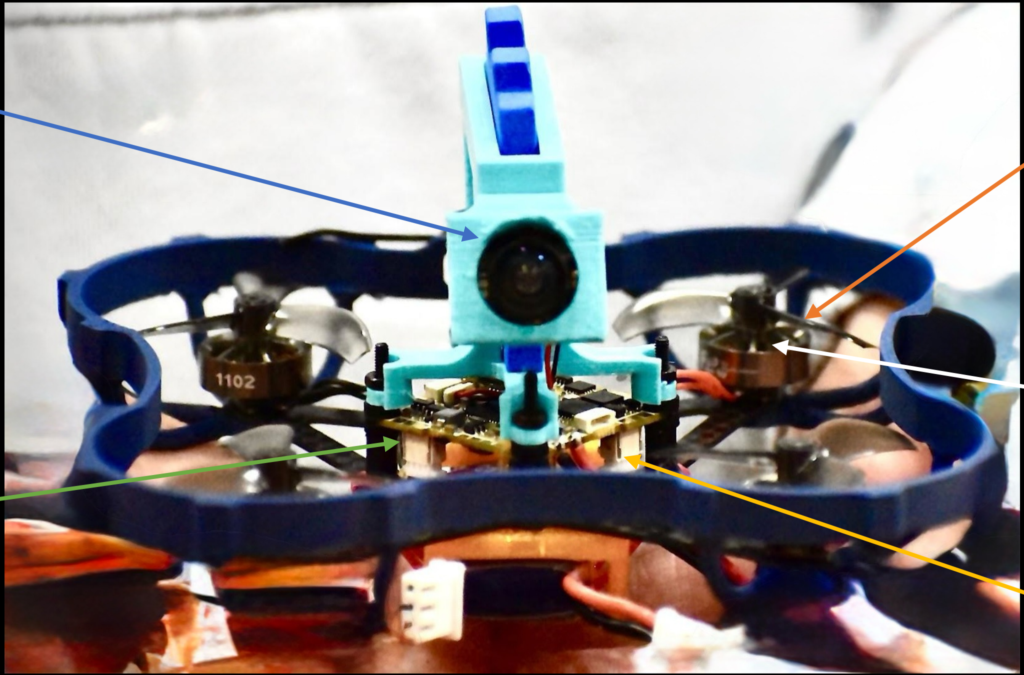
3D-printed motor guards

3D-printed prop guards

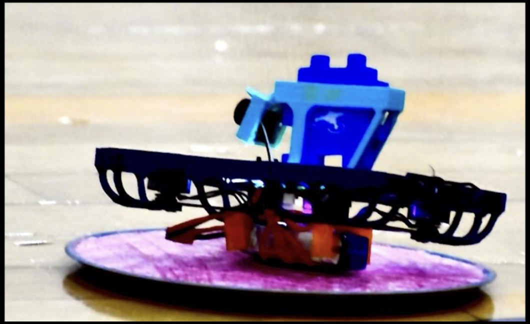
Camera/Payload mount finished part

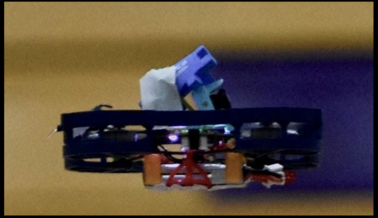
Final Assembly · COTS Parts

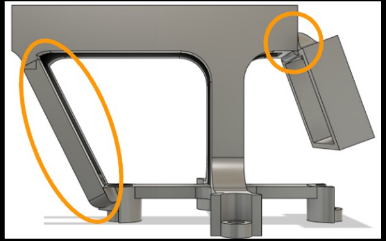




Initial Flight Testing







Upon Testing, realized the camera mount was not strong enough, so decided to add structural supports to the camera mount itself.

INTERSTELLAR FOUNDRY

Failures · Adjustments

Camera mount collapsed in flight

Vibration + thrust reversal loads not in single post design.

PID oscillations / unstable hover

ArduCopter defaults tuned for large/stiff frames; sub-300g frame has different inertia tensor.

Prop guard CF-PLA: brittle fracture in crash

CF-PLA 220N peak but catastrophic shatter; debris hazard, system unflyable.

Camera FOV missed target in terminal intercept phase

Level-mounted camera; approach from above geometry requires forward down FOV.

3-point triangulated PETG mount, silicone damping grommets.
Design for dynamic loads, not static geometry. Verify with first principles

Tuned PIDs from scratch; notch filter at 200Hz; Blackbox logging.
Never assume firmware defaults work. Tune methodically.

Tested 5 materials; selected PC (150N, plastic deformation, re-flyable).
Failure MODE > peak strength.
Ductile failure > Tensile Strength

-15 deg angled mount; recalibrated YOLO bounding box coordinates
Model engagement geometry in 3D first. Kinematic analysis before hardware.

INTERSTELLAR FOUNDRY

Detection System Architecture

01 Primary Detection

RADAR

24 GHz FMCW
K-band Doppler
Range + velocity
≤20 dBm emission

→

02 Secondary Tracking

OPTICAL

OAK-D Pro stereo
Hikvision 8MP
Angular precision
Object classification

→

03 Edge Compute

JETSON

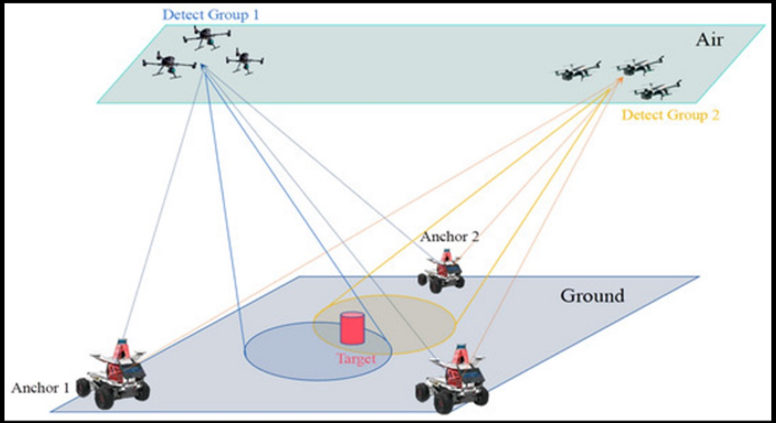
NVIDIA Jetson Nano
60+ TOPS, 7-25W
YOLOv8 inference
No cloud dependency

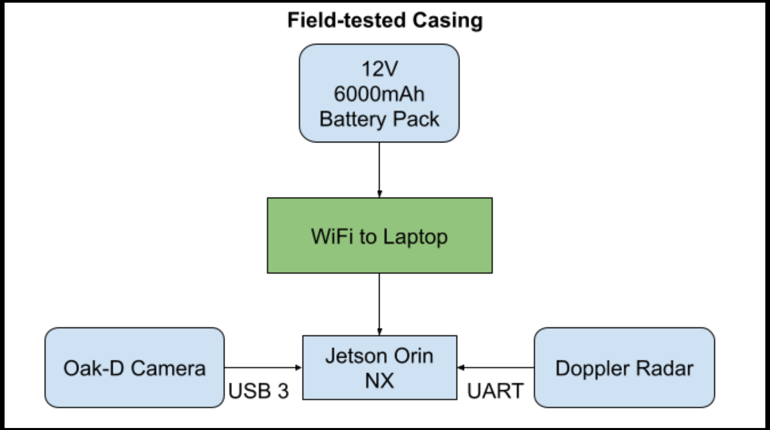
→

04 Sensor Fusion

HANDSHAKE

Kalman filter
Fuses radar + optics
Elim. false positives
Locks Group 1/2 sUAS





Sensor Fusion Physics · Kalman Filter

The math that locks Group 1/2 targets while eliminating birds and weather clutter

RADAR DETECTION

Doppler frequency shift:

```
fD = 2vr·fc / c
```

Radar range equation:

```
R = [(Pt·Gt·Gr·λ²·σ) /((4π)³·Smin)]^(1/4)
```

FMCW range resolution:

```
ΔR = c / (2·B)  where B = bandwidth
```

Velocity resolution:

```
Δv = λ / (2·T_CPI)
```

f_D: Doppler frequency shift, v_r: Relative velocity, f_c: Carrier frequency, c: Speed of light, R: Target range, P_t: Transmit power, G_t: Transmit antenna gain, G_r: Receive antenna gain, λ: Wavelength, σ: Radar cross section, S_{min}: Minimum detectable signal, ΔR: Range resolution, B: Bandwidth, Δv: Velocity resolution, T_{CPI}: Coherent processing interval

KALMAN FILTER FUSION

State prediction:

```
x^k|k-1 = F·x^k-1  +  B·uk
```

Covariance prediction:

```
Pk|k-1 = F·Pk-1·Fᵀ + Q
```

Kalman gain:

```
Kk = Pk|k-1·Hᵀ  ·  (H·Pk|k-1·Hᵀ + R)⁻¹
```

State update:

```
x^k = x^k|k-1 + Kk·(zk − H·x^k|k-1)
```

$\widehat{x_{k|k-1}}$: Predicted state estimate, F: State transition model, z_k: Measurement vector
$\widehat{x_{k-1}}$:: Previous state estimate, B: Control input model, u_k: Control vector,
$P_{k|k-1}$: Predicted covariance estimate, Q: Process noise covariance, K_k: Kalman gain,
H: Observation model, R: Measurement noise covariance, $\widehat{x_k}$: Updated state estimate

360° Azimuth Coverage

10 Hz Update Rate

<5 mrad Az/El Error

50 mph Vehicle Track Speed

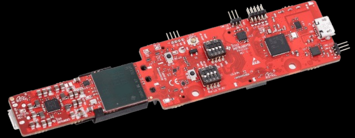
Hardware Subsystems

Radar (Tiered Scaling) ——> Optical Suite ——> Edge Processing ——> Tactical Footprint

Radar (Tiered Scaling)

↓

TI IWR6843AOPEVM (~$220) for high-speed DSP and >50m tracking.



Optical Suite

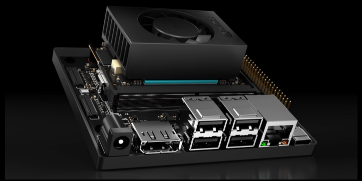
↓

Highly compatible with **OAK-D Pro** (RGB/stereo depth) or

Edge Processing

↓

NVIDIA Jetson Nano delivers localized, cloud-independent AI (60+ TOPS) on a highly efficient **7-25W power draw.**



Tactical Footprint

↓

Sub-10 lb modular design supports under 15-minute deployment on ISV, JLTV, FMTV, and HEMTT platforms.

Software Subsystems

GNSS-Denied Navigation ——> Spectral Stealth ——> Cybersecurity

GNSS-Denied Navigation

↓

Converts 2D pixels to 3D map coordinates using OpenCV and SLAM to localize threats without GPS.

Spectral Stealth

↓

Air-gapped serial communication and passive optics ensure maximum EMI robustness and low physical/spectral signatures.

Cybersecurity

↓

Architecture strictly complies with **DOD Instruction 8510.01 (RMF)** to streamline the Authorization to Operate (ATO).

LOE 1 Homeland	ASCII output · FAAD C2 · 5-30kW gen compatible
LOE 2 Tactical/Mobile	GNSS-denied · ISV/JLTV mount · passive posture

Engagement Logic · Swarm Software

GROUND STATION SOFTWARE

Python / OpenCV

Real-time video processing pipeline

HB100 Doppler

Velocity + range primary detection

Kalman (FilterPy)

Multi-target tracking + fusion

YOLOv8 inference

Jetson-accelerated classification

PyQt6 UI

Operator display / engagement cue

ENGAGEMENT LOGIC

Target acquired

Dual-sensor confirm → lock

Primary dispatch

Lead interceptor → head-on vector

Flankers launch

2 flankers on diverging intercept

Abort logic

Flankers abort if primary scores kill

Re-engage

Missed intercept → re-cue loop

MVP SUCCESS CRITERIA

Dual-sensor detect

Both radar AND camera confirm

80+ km/h targets

Intercept at full speed

3 simultaneous launches

Coordinated swarm execution

Kinetic contact

Physical intercept confirmed

< 15 min deploy

From packed to combat ready

Business Case · Current Market

The gap: low-cost, high-lethal, mass-producible C-UAS at scale

TIERED ROM PRICING

TIER 1 1-5 units

$500-$1,000/node

Prototype / evaluation

TIER 2 6-25 units

LRIP pricing

Low-rate initial production

TIER 3 26-100+ units

Mass manufacturing

Full contract scale

MARKET CONTEXT

$4B+

C-UAS global market by 2030

~$500K

Avg. legacy sensor node cost

<$1K

Our node cost is 500× cheaper

Group 1/2

Primary threat is underserved

Parth Patel

Interstellarfoundry.com

linkedin.com/company/interstellarfoundry

@ ppatel432@gatech.edu